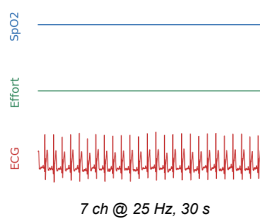
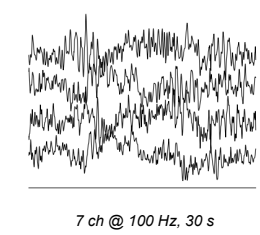


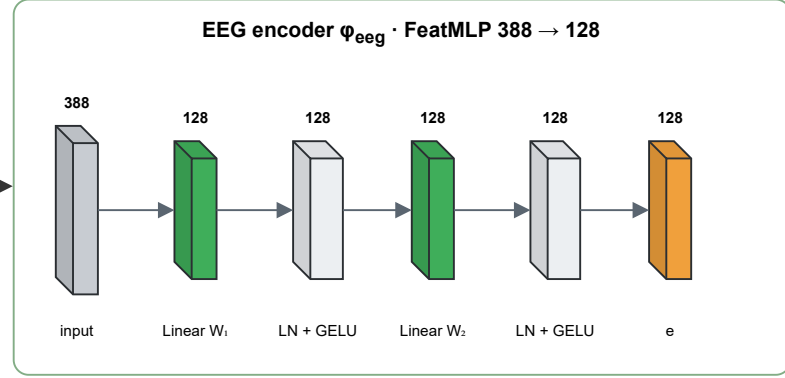
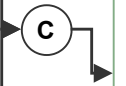
Two-stream multimodal multi-task network

$$f_{\theta} : (f^{\text{eeg}}, z^{\text{fnd}}, x^{\text{car}}) \rightarrow (\hat{y}^{\text{stg}}, \hat{y}^{\text{apn}})$$

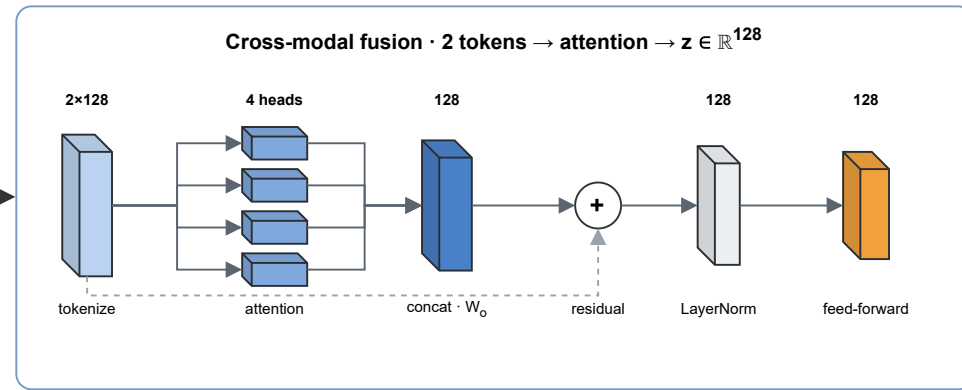
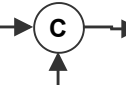
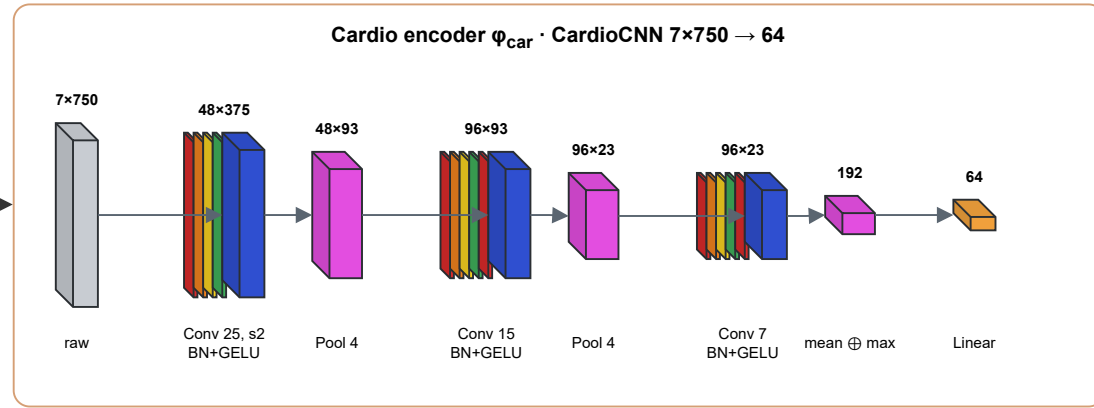


Expert physiology
 $f^{\text{eeg}} \in \mathbb{R}^{188}$
 band power, spindle, Hjorth

Frozen LaBraM *
 $z^{\text{fnd}} \in \mathbb{R}^{200}$
 4 EEG channels, no gradient



Raw cardiorespiratory tensor
 $x^{\text{car}} \in \mathbb{R}^{7 \times 750}$
 per-subject z-score



Staging head
 $512 \rightarrow 5$
 + HMM Viterbi decode

BiLSTM 2 layers, bidirectional
 context L = 20 epochs (10 min)
 $h_t \in \mathbb{R}^{512}$

Respiratory head
 $[h_t ; c_t] \cdot 576 \rightarrow 256 \rightarrow 1$
 direct cardio bypass



dashed outline = frozen (no gradient) · 2,764,774 trainable + 5,819,936 frozen parameters